Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 87899

Part A: Each question is worth 3 points

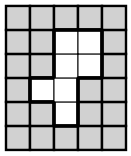
1. A book has 104 pages, numbered from 1 to 104. Mia writes all the page numbers on cards, one digit on each card.
How many cards does Mia need?

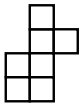
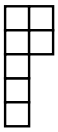
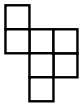
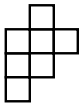
A. 312 ***B.*** 213 ***C.*** 208 ***D.*** 204 ***E.*** 104

2. The odometer of a car shows 458937 km. All the digits of this number are different.
After how many more kilometres will all the digits be different again?

A. 24 ***B.*** 23 ***C.*** 22 ***D.*** 13 ***E.*** 1

3. Which piece covers the empty area exactly? The pieces may be turned and flipped over.



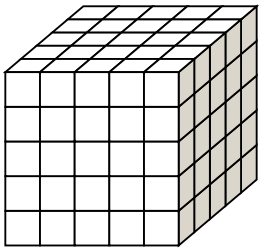
A.  ***B.***  ***C.***  ***D.***  ***E.***

Part B: Each question is worth 4 points

4. Chloe, Emma, Ben, Hana and Ravi were born in different months: June, December, March, July and October. Chloe and Hana both were born in a month that starts with J. Ben was born in winter. Chloe was not born in June. Ravi was born in spring.
Which statement is true?

A. Ben was born in December ***B.*** Hana was born in October ***C.*** Chloe was born in December
D. Ben was born in March ***E.*** Ravi was born in July

5. Mia glued 125 small cubes into a big 5 × 5 × 5 cube (like the picture) and painted the whole outside of it, also the bottom. Then the big cube falls apart.



How many small cubes have paint on exactly one face?

A. 12 ***B.*** 54 ***C.*** 58 ***D.*** 61 ***E.*** 64

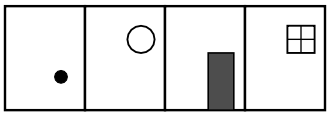
6. Omar was born on February 1, 2016 and Leo on November 21, 2017.
How many years old is Leo on October 15, 2024?

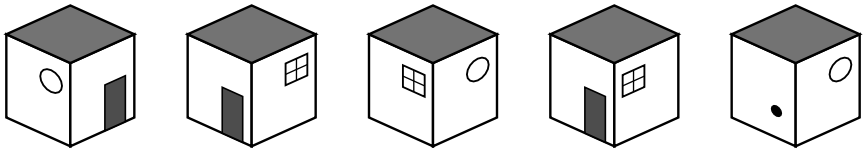
A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

7. Finn's snake toy is 16 dm 3 cm long. Dan's snake toy is 1120 mm long.
How many centimetres longer is Finn's snake?

A. 40 cm ***B.*** 41 cm ***C.*** 42 cm ***D.*** 51 cm ***E.*** 54 cm

8. The strip below is cut out and folded into the four walls of a house, with the pictures on the outside. A flat roof is put on top.



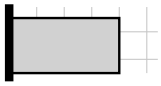


A B C D E

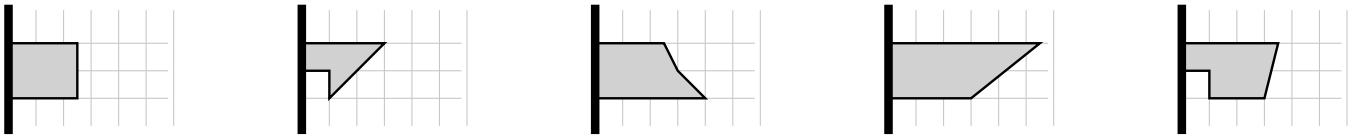
How many of the houses A–E can be made from the strip?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 4

9. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Sara draws the shadow at five different moments. Flat, the flag is 4 squares wide and 2 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat



A B C D E

How many of the five drawings can be right?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

10. What is the smallest number whose digits add up to 28?

A. 9991 ***B.*** 2989 ***C.*** 2899 ***D.*** 1999 ***E.*** 1605

11. Which word has exactly 1 different letter in common with NUMBER?

A. KANGAROO ***B.*** BUTTERFLY ***C.*** DINOSAUR ***D.*** PROBLEM ***E.*** WINDOW

Part C: Each question is worth 5 points

12. A bar code is made of 22 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 1 more white bars than wide black bars.

How many narrow black bars are there?

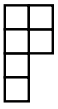
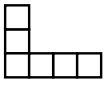
A. 10 ***B.*** 9 ***C.*** 3 ***D.*** 1 ***E.*** 0

13. Martha has the cards 1, 5, 7 and 8. Using every card exactly once, with +, –, ×, ÷ and brackets, Martha wants to get the biggest possible number.

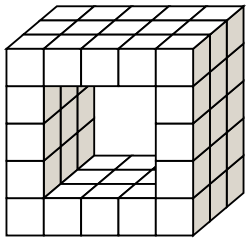
What is the biggest number Martha can get?

A. 21 ***B.*** 279 ***C.*** 280 ***D.*** 315 ***E.*** 336

14. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A. ***B.*** ***C.*** ***D.***  ***E.*** 

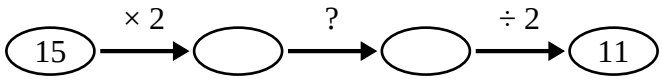
15. Mia's tunnel is 3 cubes long. The hole goes straight through. Mia makes it longer by adding more cubes at the back in the same pattern, until it is 7 cubes long.



How many cubes are in the long tunnel?

A. 112 ***B.*** 128 ***C.*** 129 ***D.*** 139 ***E.*** 175

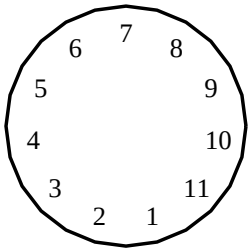
16. One arrow has lost its calculation. The others are done one after the other.



Which calculation belongs on the arrow with the ?

A. + 6 ***B.*** – 8 ***C.*** × 2 ***D.*** × 3 ***E.*** + 8

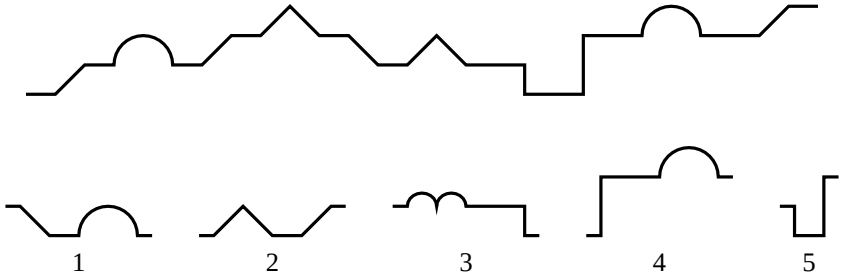
17. A round cookie with the numbers 1 to 11 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



Where does the crack go?

A. between 8 and 9, and between 2 and 3
B. between 9 and 10, and between 2 and 3
C. between 7 and 8, and between 3 and 4
D. between 7 and 8, and between 1 and 2
E. between 6 and 7, and between 11 and 1

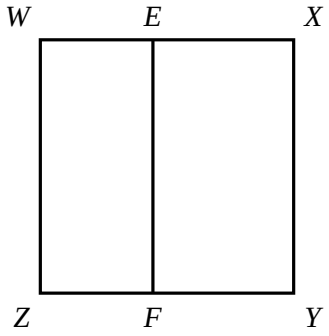
18. Lena cut a strip of paper along this line.



How many of the pieces 1–5 below belong to the line?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 5

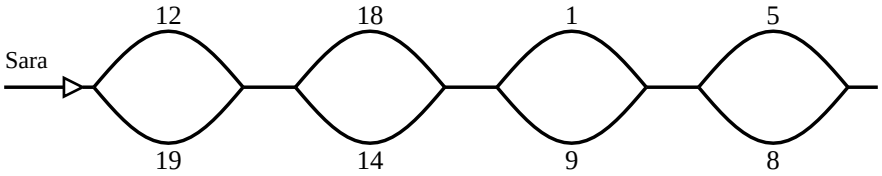
19. The square *WXYZ* is cut into two rectangles. The perimeter of *WEFZ* is 26 cm and the perimeter of *EXYF* is 28 cm.



What is the perimeter of the square *WXYZ*?

A. 36 cm ***B.*** 37 cm ***C.*** 44 cm ***D.*** 45 cm ***E.*** 54 cm

20. Sara walks along the road from left to right and adds up the numbers along the way. The total is 42.



Which number did Sara see at the second fork?

A. 1 ***B.*** 12 ***C.*** 14 ***D.*** 18 ***E.*** 19

Answer Key

1.	**D**	**2.**	**B**	**3.**	**A**	**4.**	**A**	**5.**	**B**	**6.**	**A**
7.	**D**	**8.**	**D**	**9.**	**B**	**10.**	**D**	**11.**	**E**	**12.**	**D**
13.	**E**	**14.**	**D**	**15.**	**A**	**16.**	**B**	**17.**	**A**	**18.**	**B**
19.	**A**	**20.**	**C**								

Solutions

1. D (204) — 9 × 1 + 90 × 2 + 5 × 3 = 204. [2003 Q16 · digits_for_pages]

2. B (23) — Counting up, every number before 458960 repeats a digit. 458960 has all different digits: 458960 – 458937 = 23. [2002 Q24 · all_different]

3. A — The empty area has 7 squares; all pieces do too, so look at the shape. Turn piece A and it fits exactly; each of the others has one square in a different place. [2003 Q18 · one_piece]

4. A (Ben was born in December) — The only possible matching: Chloe – July; Emma – October; Ben – December; Hana – June; Ravi – March. [2002 Q18 · which_true]

5. B (54) — They are in the middle of each face: 6 × 3 × 3 = 54. [2003 Q9 · painted]

6. A (6) — 2024 – 2017 = 7, but Leo's birthday (November 21) has not come yet that year, so Leo is 6. [2007 Q12 · age_on_day]

7. D (51 cm) — 16 dm 3 cm = 163 cm and 1120 mm = 112 cm. 163 – 112 = 51 cm. [2003 Q10 · how_much_longer]

8. D (3) — A house is right if its two walls are neighbours on the strip, in the same order, with the marks on the correct side. Right: A, B, E, so 3. [2003 Q12 · how_many_houses]

9. B (2) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: B, D, E. So 2 can be right. [2002 Q15 · count_possible]

10. D (1999) — Use as few digits as possible: as many 9s as possible, with the smallest digit in front. 1999: 1 + 9 + 9 + 9 = 28. [2007 Q4 · smallest_with_sum]

11. E (WINDOW) — WINDOW shares N with NUMBER. KANGAROO shares 2; DINOSAUR shares 3; BUTTERFLY shares 4; PROBLEM shares 4. [2007 Q3 · which_word]

12. D (1) — Starting black and ending white, the bars come in pairs: 11 black and 11 white. Wide: 11 – 1 = 10. Narrow: 11 – 10 = 1. [2003 Q17 · ends_differ]

13. E (336) — (1 + 5) × 7 × 8 = 336. Multiplying by 1 changes nothing, so it is better to add the 1 to the smallest other card first. [2002 Q21 · biggest]

14. D — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece D fit together like that. [2007 Q8 · two_copies]

15. A (112) — Each slice of the tunnel has 5 × 5 – 3 × 3 = 16 cubes. 7 slices: 16 × 7 = 112. [2002 Q20 · longer_tunnel]

16. B (– 8) — Before that arrow we have 30; working back from 11, after it we need 22. Only – 8 turns 30 into 22. [2007 Q5 · which_operation]

17. A (between 8 and 9, and between 2 and 3) — Total 66, so each piece has 33: 9+10+11+1+2 = 33 and 3+4+5+6+7+8 = 33. [2002 Q11 · straight_crack]

18. B (1) — Only piece 4 fits. Not: 1 (it has a round dome where the line has a pointed roof); 2 (it has the right shapes but in the wrong order); 3 (it has two small bumps where the line has a pointed roof); 5 (its flat part between the two features has the wrong length). [2002 Q7 · how_many_fit]

19. A (36 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 26 + 28 = 54 cm = 6 sides. Side = 54 ÷ 6 = 9 cm, perimeter 4 × 9 = 36 cm. [2002 Q6 · two_perimeters]

20. C (14) — Only one route gives 42: 19 + 14 + 1 + 8 = 42. At the second fork: 14. [2007 Q1 · which_branch]